Muhammad Taha Irfan

Islamabad, Pakistan | +923180574423 | tahairfan549@gmail.com

Social networks

- https://www.linkedin.com/in/taha-irfan05/
- https://github.com/Taha-tech05
- https://my-portfolio-nine-omega-35.vercel.app/

Summary

Results-driven Computer Science student and Full-Stack Developer with a track record of architecting scalable web applications using React.js, Next.js, and Node.js alongside deep proficiency in PostgreSQL, C++, and Python. Expert in the Machine Learning lifecycle and AI-driven solutions, with hands-on experience in data preprocessing, model fine-tuning/evaluation, and the integration of intelligent agents into high-performance systems. A proactive "builder" adept at merging 3D visualizations with clean code to solve complex, unstructured problems through innovative engineering and collaborative research.

Experience

Software Developer **09/2026 to 11/2026**
Robotmea Edutech Pvt. Ltd. Islamabad, Pakistan

- Tested the Campus Management System (CMS) end to end across its admissions, HR, finance, and assessment modules, logging and resolving defects before release.
- Deployed the CMS to a live environment and supported the rollout, fixing issues reported after go-live.
- Designed and shipped new features and workflow improvements driven directly by user feedback and demand from school staff.

Research Assistant – AI-Enhanced Digital Math Visualization for Gender Parity in STEM **03/2026 to 03/2027**
FAST-NUCES (FRSG) Islamabad, Pakistan

- Building a Windows-based offline 3D educational platform using React Three Fiber, with a Student Mode (interactive parametric shapes) and a Teacher Mode (session-analytics dashboard), enabling 200+ students to interact through real-time manipulation and behavioral tracking.
- Engineered a rule-based inference engine analyzing 8+ behavioral metrics (slider usage, rotation patterns, inactivity duration, extreme value exploration) to trigger adaptive hints, improving student engagement by $\sim$30% across 5+ modules.
- Designed bilingual UI/UX (English/Urdu) with React components, ensuring accessibility and reducing onboarding time by 40%; followed agile development cycles with iterative feedback integration.
- Application development is complete; next phases are classroom deployment to 100+ students and collection/analysis of behavioral datasets for a planned co-authored research paper on adaptive scaffolding.

Software Development Intern **06/2026 to 08/2026**
Robotmea Edutech Pvt. Ltd. Islamabad, Pakistan

- Contributed to a Campus Management System (CMS) modernizing a legacy school portal into a full platform (React, Express, PostgreSQL/Prisma, Socket.IO) intended to support 1,000+ students across multiple partner schools.
- Built modules for Admissions, HR, Financials (double-entry ledger, payroll), Certification (PDF generation), and academic assessment/reporting.

- Diagnosed and fixed an N+1 query bottleneck in payroll/fee generation via batching, reducing the projected full-batch run time for 1,000+ students to under 2 seconds.

AI Intern **06/2025 to 08/2025**
AI Genmat Islamabad, Pakistan

- Developed and evaluated ML models for real-world classification tasks.
- Automated end-to-end data collection via BeautifulSoup and Selenium, reducing manual acquisition time significantly.
- Engineered preprocessing pipelines (missing values, outlier removal, feature engineering) using Pandas and Scikit-learn, improving dataset quality for downstream model training.

Education

Bachelor of Science: Computer Science **2027**
FAST NUCES Islamabad, Pakistan

- Year of start: 2023
- **CGPA: 3.73**
- Completed rigorous coursework in Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Web Programming, Database Systems, Artificial Intelligence, Operating Systems, High Performance Computing with GPUs, Computer Networks, and Generative AI.
- Gained hands-on experience in C++, C#, Python, Java, SQL, React.js, and Next.js.
- Worked on different Projects in this tenure.

FSc in PreEngineering **2023**
Scholars Science College Wah Cantt, Pakistan

- Grade: A
- Built a solid foundation in core subjects including Mathematics, Physics, and Chemistry, developing strong analytical and problem-solving skills.
- Maintained consistent academic performance and prepared extensively for entry into top-tier engineering and computer science programs.
- Developed discipline and a strong work ethic through a rigorous study schedule and exam preparation.

Matric in Computer Science **2021**
Sir Syed School Wah Cantt, Pakistan

- Grade: A
- Studied core science subjects including Computer Science, Physics, Mathematics, and Chemistry, laying the groundwork for a career in technology.

Skills

- Multitasking Efficiency
- Problem Resolution
- Project Leadership and Ownership
- Machine Learning Model Testing
- Data Preprocessing and Cleaning
- Transformer Fine-Tuning
- AI Agent Integration
- Agile / Scrum
- Entity Framework
- Prompt Engineering
- Codex, Antigravity and Claude Code
- Debugging & Performance Optimization

- Full-Stack Web Development
- HTML5, CSS3, and JavaScript (ES6+)
- React.js and Next.js
- Angular
- Node.js and Express
- C#, and .NET
- PostgreSQL and MongoDB
- C++, Python, and Java Programming
- REST APIs
- Git, GitHub, Linux, and Socket.io
- Pandas, NumPy, and Scikit-learn

Projects

1. ShifaKinetix – AI-Governed Musculoskeletal Triage & Consultation Platform (Final Year Project) (Hard) | Technologies: 3D Anatomical Model, Multi-Agent AI, Telehealth
Tenure: Approved – In Development
Proposed a patient-doctor telehealth platform using a segmented 3D anatomical model for symptom reporting, with a deterministic red-flag safety gate preceding any AI involvement. Scope covers a governed multi-agent pipeline (triage, consultation, report review) with mandatory doctor sign-off on all AI-generated clinical output. Architecture reviewed and approved by the FYP panel; currently at data-collection and design stage, with implementation in the final two semesters.

2. Transformer-Based Clinical Trial Eligibility Matching System (Hard) | Technologies: PyTorch, Transformers, LoRA, Python
Tenure: October – November 2026
Fine-tuned an open-weight decoder Transformer on the TrialGPT Criterion Annotations dataset to match patient descriptions against clinical trial inclusion/exclusion criteria with evidence-grounded explanations. Implemented KV-cached autoregressive decoding, tiled online-softmax attention, and greedy speculative decoding from scratch, each verified against reference implementations for numerical/token-level correctness. Built a browser-based application exposing criterion-level eligibility results, supporting evidence, and inference-mode benchmarking.

3. Multi-Task Generative AI Suite (Hard) | Technologies: React, FastAPI, ONNX, Optuna, Docker
Designed and deployed four generative-AI systems (universal denoising autoencoder, classifier-routed specialist autoencoders, a jointly trained soft mixture-of-experts restorer, and a conditional GAN for face-to-sketch generation) unified in one React/FastAPI application with ONNX inference, Optuna-tuned hyperparameters, and Docker deployment.

4. AirControlX – Automated ATC System (Hard) | Language: C/C++, SFML
Tenure: April – May 2025
Simulated real-time air traffic management using multithreading and process scheduling. Developed a graphical simulation to visualize flight paths, analytics, and violation handling protocols.

5. GitLite – Version Control System (Hard) | Language: C++
Tenure: December 2024 – January 2025
Designed a custom version control system optimized for large files. Implemented Red-Black Trees and Merkle Trees to ensure data integrity and efficient branching, merging, and versioning.

6. TravelEase – Travel Management System (Hard) | Technologies: C#, .NET, SQL Server
Tenure: April – May 2025
Built a multi-role desktop application for bookings and payments. Features role-based access for admins, travelers, and service providers, backed by a robust SQL schema and professional WinForms GUI.

7. Intelligent Reading Comprehension System (Hard) | Technologies: Python, Scikit-Learn, Streamlit
Tenure: April-May 2026
Developed an AI pipeline for quiz generation using the RACE dataset ($\sim$100K questions). Built Model A for answer verification using Logistic Regression and SVM, integrated into a 4-screen Streamlit UI.

8. Property Dealer CRM System (Hard) | Technologies: Next.js, Node.js, PostgreSQL
Tenure: April – May 2026
Live: `https://propertycrm.up.railway.app/`
Architected a production-ready CRM with JWT/NextAuth authentication. Implemented real-time lead notifications via Socket.io and a rule-based lead scoring system across three priority tiers.

9. AI-Powered E-commerce Listing Platform (Hard) | Technologies: FastAPI, Gemini API, PostgreSQL
Tenure: February – April 2026

Developed a whole dashboard. Connected a JS frontend to a FastAPI backend to automate product listings for Amazon and Etsy. Created tables for listings and connected database with backend.

10. Financial Tracking System (Hard) | Technologies: Next.js, Node.js, PostgreSQL
Tenure: April – May 2026
Live: `https://financial-tracker-donation-manage.vercel.app`
Built a full-stack finance management MVP: automated pledge vs. received donation tracking, real-time project balance calculations, and RBAC with JWT authentication.

11. Rentify – Rental Management System (Hard) | Technologies: JavaFX, SQL Server
Tenure: January – April 2026
Collaborated on a complete rental ecosystem. Implemented the full lifecycle of rental activity, including product listing, automated return flows, and a mutual rate-and-review system between owners and renters.

12. Smart Drone Simulation (Hard) | Technologies: C++, A* Algorithm
Tenure: June-July 2025
Developed a navigation system using the A-Star (A)* AI algorithm. Implemented custom grid logic with variable terrain costs (Windy vs. Common) to calculate the most efficient and safe path to a goal.

13. CPU Parallel Reduction Optimization (Hard) | Technologies: C++, OpenMP, AVX2 Intrinsics
Tenure: April – May 2026
Developed a 5-step optimization progression for integer sum reduction as a counterpart to Mark Harris's GPU study. Implemented manual AVX2 vectorization and false-sharing remediation, reaching 97.6% of the STREAM dual-channel bandwidth ceiling for cosine similarity workloads. Evaluated performance systematically across the cache hierarchy to identify optimal implementation regimes.

14. Centipede – Classic Arcade Game (Medium) | Language: C++, SFML Library
Tenure: November-December 2023
Recreated the vintage arcade experience with keyboard-driven controls. Focused on implementing collision detection, scoring systems, and sprite-based animations for enemy movement.

15. Plants vs. Zombies Clone (Medium) | Language: C++, SFML Library
Tenure: April-May 2024
Developed a grid-based tower defense game. Engineered core game mechanics including plant-placing logic, enemy wave spawning, and real-time graphics rendering for a smooth gameplay experience.

16. 2D Cricket Batting Game (Medium) | Technologies: React.js, JavaScript, HTML5
Tenure: February 2026
Engineered a probability-based sports game using React hooks. Implemented a dynamic power-bar with click-based shot detection and synchronized DOM animations, mapping outcomes strictly to probability regions.

17. House Price Prediction Model (Medium) | Technologies: Python, Scikit-learn, Pandas
Tenure: July – August 2025
Implemented a Linear Regression model with Gradient Descent from scratch. Applied IQR-based outlier removal and one-hot encoding on the Ames Housing dataset to achieve an R^2 of **0.805**.

18. Large-Scale Web Scraper (Medium) | Technologies: Python, BeautifulSoup, Selenium
Tenure: July – August 2025
Built a reusable data pipeline to scrape and clean 1,500+ company profiles. Integrated regex-based validation and deduplication logic to ensure high data integrity for export.

19. NoteVault – Advanced Note Management (Medium) | Technology: React.js
Tenure: January 2026
Live: `https://notevault-three.vercel.app/`
Deployed a comprehensive cloud-based note-taking application. Focused on high-performance rendering, secure data persistence, and a sleek, responsive user interface for modern productivity.

20. Console-Based Notepad (Easy) | Language: C++
Tenure: September – October 2024
Built a terminal-based text editor focusing on file I/O operations. Implemented essential features including file read/write, real-time editing, and save functionality using low-level keyboard input handling.

21. Newsletter Subscription System (Easy) | Technologies: Node.js, Express.js, Mailchimp API
Tenure: January 2025
Developed a functional email signup platform. Integrated the Mailchimp API to manage subscriber data, focusing on backend form validation and providing real-time user feedback.

22. Clothing E-commerce Website (Easy) | Technologies: HTML, CSS, Bootstrap
Tenure: November 2023
Designed a responsive static fashion website. Utilized Bootstrap components to create a clean UI featuring product categories and mobile-friendly layouts.

23. Blog Web Application (Easy) | Technologies: HTML, CSS, JavaScript, Node.js, Express.js
Tenure: February 2025
Created a full-stack blog platform supporting basic CRUD operations (Create, Read, Update, Delete). Implemented dynamic routing and templating to manage posts and user interactions.

Hobbies and interests

- Cricket is my passion, and I like to both play and watch cricket
- I occasionally watch tennis, football, and table tennis
- I am passionate about making productive applications and solving people's problems.
- I am interested in software development and AI engineering.

Languages

Urdu: C2
Proficient

English: B2
Upper Intermediate

Punjabi: B2
Upper Intermediate

Hindi: B2
Upper Intermediate